Shanghai Pinghe School, High School Dept. IEP

Name:_____ Class:_____

First Monthly Exam （Version A）

Autumn Semester 2024-2025

G12 Post AP Biology: Neuroscience

Duration: 60 minutes

INSTRUCTION

✧ Do not open the exam paper until you are instructed to do so.

✧ Answer all questions on the paper provided.

✧ The full mark for this exam is **120 marks.**

✧ Do not hand- in the paper before the end of the exam.

✧ The total exam paper contains **8 Pages**

Part I: Single-Choice Questions **[2*15=30 marks]**

1. How does tetrodotoxin (TTX) affect channels?

A. TTX blocks K+ channels.
B. TTX blocks Na+ channels.
C. TTX blocks Cl– channels.
D. TTX sensitizes voltage-gated Na+ channels.

2. The movement of what ion occurs in the rising phase of the action potential?

A. Inward Na+
B. Outward K+
C. Outward Na+
D. Inward K+

3. How long does an action potential last from the beginning of the rising phase to the end of the falling phase?

A. 2 nanoseconds
B. 2 msec
C. 2 seconds
D. 2 microseconds

AKRD

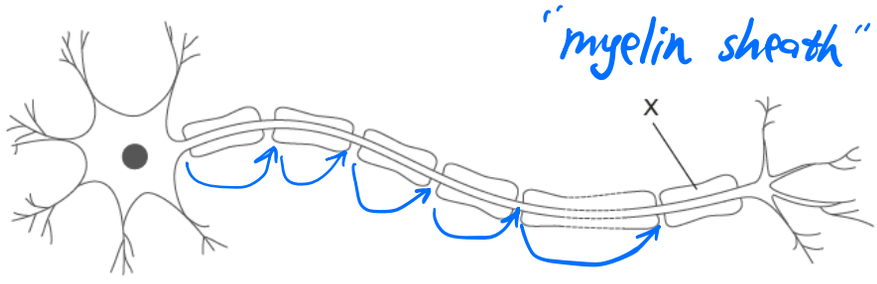
4. Identify the protein that helps anterograde transport move materials from the soma to the terminal.

A. Dynein
B. Kinesin
C. Pyruvic acid
D. MAP

5. A neuroscientist is investigating how different neural circuits in the brain analyze sensory information, form perceptions of the external world, make decisions, and execute movements. At what level of analysis is this research conducted? Choose the correct option.

A. Molecular neuroscience level
B. Cellular neuroscience level
C. Systems neuroscience level
D. Cognitive neuroscience level

6. The image shows a neuron.



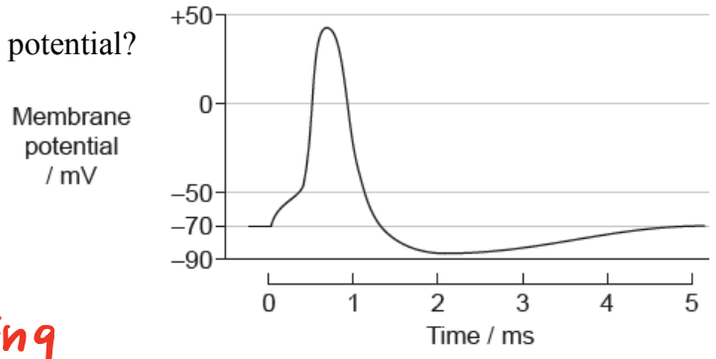
What is the function of X?

A. Increases the speed of transmission along the axon
B. Increases the rate of exchange of sodium and potassium ions
C. Holds bundles of neurons together to form a nerve
D. Determines the direction of the action potential

7. The graph shows changes in the membrane potential in an action potential.

What is the approximate value of the threshold potential?

A. −88 mV
B. −70 mV
C. −50 mV
D. +45 mV



K+ leaking

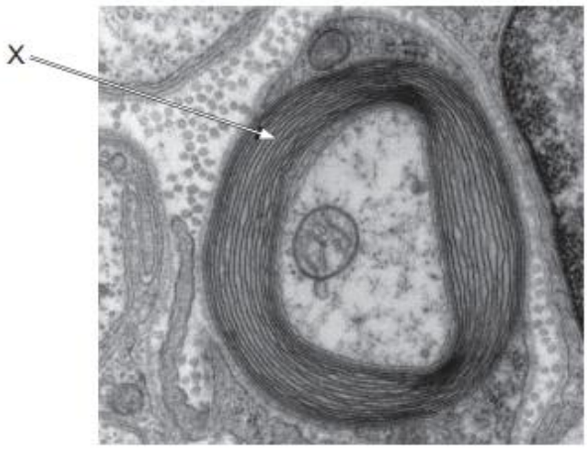
8. Which statement applies to an axon at rest? "内负外正" RP=-65mV

A. There is no electric potential difference between the external and internal surfaces of the plasma membrane.
B. The external surface of the plasma membrane is positive relative to the internal surface.
C. The external surface of the plasma membrane is negative relative to the internal surface.
D. The internal surface of the plasma membrane has a much higher concentration of sodium ions.

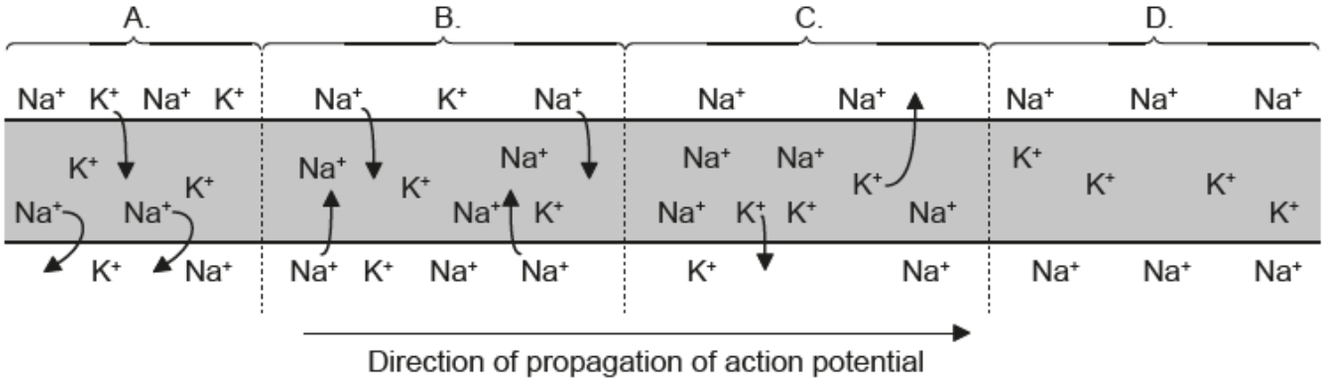
9. The electron micrograph shows a transverse section through a myelinated neuron.

What process is facilitated by the presence of the structure labelled X?

A. Repolarization of the nerve cell membrane
B. Generation of an action potential
C. Saltatory conduction
D. Synaptic transmission



10. The diagram shows an action potential moving along a neuron. Which part of the diagram represents depolarization?



11. How does potassium move across the membrane of a neuron during repolarization?

A. Simple diffusion
B. Facilitated diffusion
C. Endocytosis
D. Active transport

-50 → -70

AP

12. Neurons transmit electrical impulses. Which statement describes part of this process?

A. K+ ions are pumped out of the cell to depolarize the membrane.
B. Ion channels let K+ diffuse into the cell to depolarize the membrane.
C. Na+ ions are pumped into the cell to repolarize the membrane.
D. Ion channels let Na+ diffuse into the cell to depolarize the membrane.

13. The central sulcus separates which lobes?

A. Temporal from parietal
B. Frontal from parietal
C. Frontal from temporal
D. Parietal from occipital

} Unit 9.

14. Because the resting potential is _____, depolarization means _____ while hyperpolarization means _____.

A. negative, more positive, more negative
B. negative, more negative, more positive
C. positive, more positive, more negative
D. positive, more negative, more positive.

AP

15. the maximum firing rate of a neuron is/are determined by

A. relative refractory period
B. inactivation period of Na+ channel.
C. inactivation period of K+ channel.
D. threshold potential

Part II: Matchup [20 marks]

Matchup the following individuals to their findings / views towards brain function. **[10 marks]**

16. Broca
17. Flourens
18. Galen
19. Aristotle
20. Hippocrates

A. -brain was involved in sensation and the seat of intelligence

B. -the heart was the center of intellect
-brain was a radiator for cooling blood that was overheated.

C. -Distinguish cerebrum & cerebellum by sheep brain dissections

D. -most vociferous critics of phrenology
-performed experimental ablations of brain

E. -found a brain area in the left frontal lobe responsible to produce speech.

Matchup the following disorder to the description. **[10 marks]**

21. Stroke
22. Parkinson's disease
23. Cerebral palsy
24. Alzheimer's disease
25. Spinal paralysis

A. A progressive disease of the brain that leads to difficulty in initiating voluntary movement

B. A loss of feeling and movement caused by traumatic damage to the spinal cord

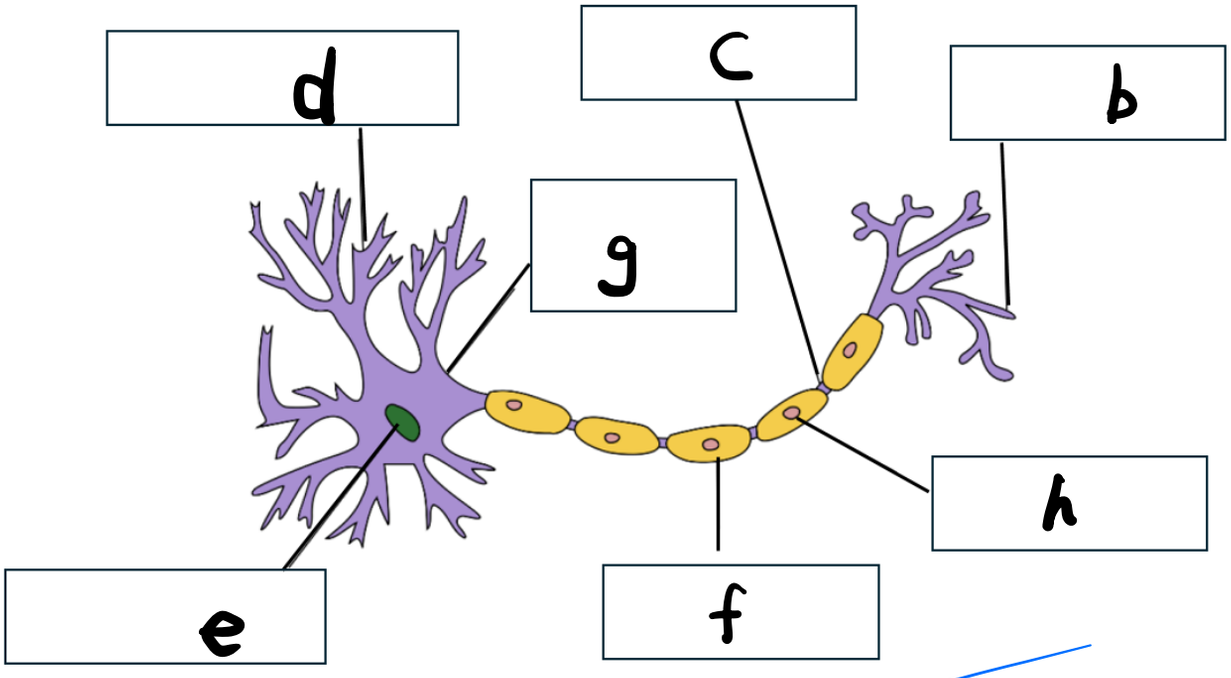
C. A loss of brain function caused by disruption of the blood supply, usually leading to permanent sensory, motor, or cognitive deficit.

D. A progressive degenerative disease of the brain, characterized by dementia and always fatal

E. A motor disorder caused by damage to the cerebrum before, during, or soon after birth

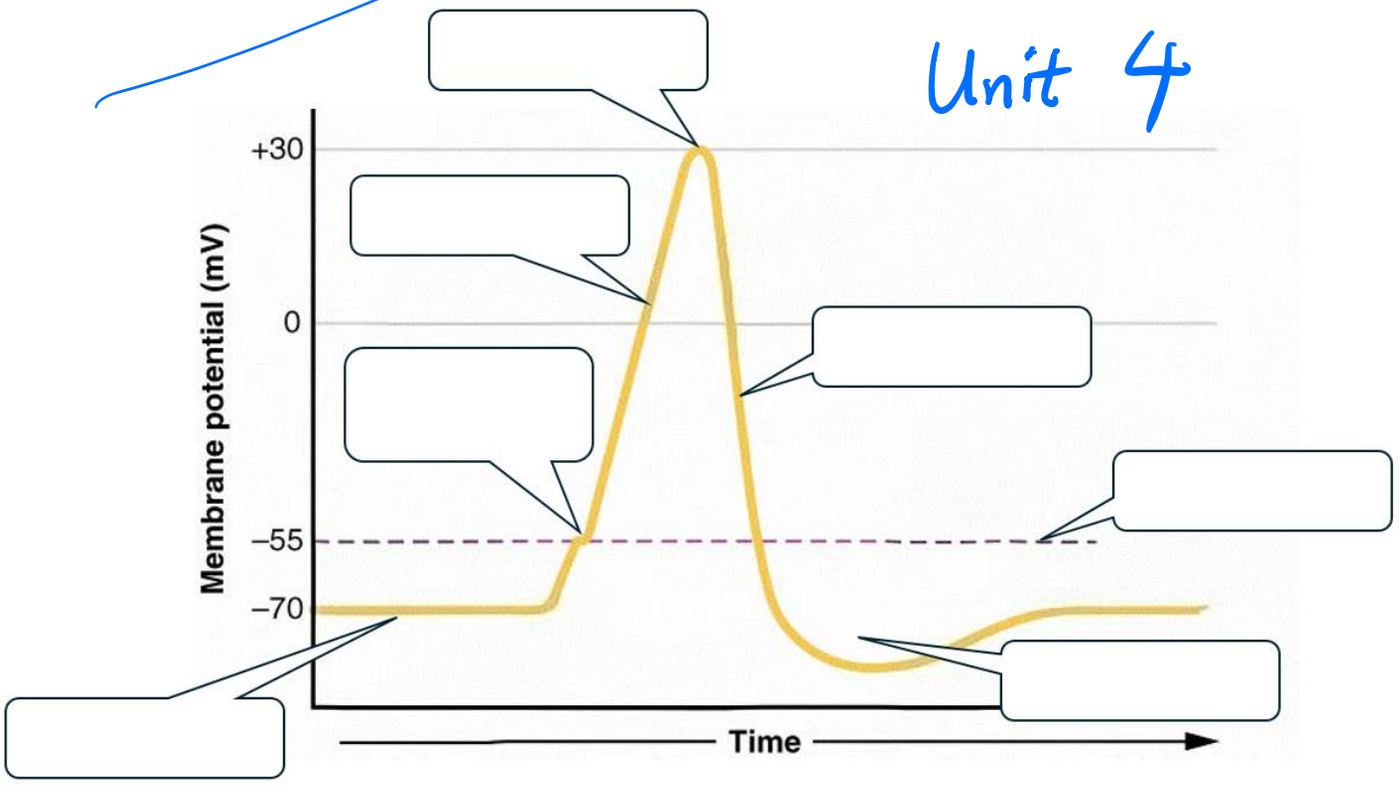
Part III: Fill in the blanks using the correct terms from the table. [28 marks]

a. Dendritic spine	b. Axon terminal	c. Node of Ranvier	d Dendrite
e Soma	f. Myelin Sheath	g. Axon Hillock	h. Nucleus of Schwann Cell



Voltage-gated Na+ channel Open	depolarization	threshold potential	Hyperpolarization
Voltage-gated K+ channel Open	repolarization	resting potential	

Unit 4



Part IV: Free Response Questions **6 marks * 5 =30 marks**

Chose 5 of the following questions

	O	I
Na^+	150	15
K^+	5	100

COPS

1. Distinguish between **Nissl Stain** Method and **Golgi Stain** Method.
2. State **TWO** different method of speed up propagation of nerve impulse.
3. Compare and Contrast **Oligodendroglia** and **Schwann Cell**
4. State the Intracellular & Extracellular $\mathbf{K^+}$ and $\mathbf{Na^+}$ concentration gradient in a typical neuron.
5. Using **TWO** named examples of glia cell, state their specific functions.
6. Structure features reflect functional roles, state the roles of **dendrite, soma, dendritic spine** and **axon** in nerve signaling.
7. State **TWO** mechanisms of regulating external potassium concentration in the brain .
8. State **TWO** difference between intracellular recording and extracellular recording.
9. **Explain** what is meant by 'All- or- None' features.
10. **State** the specific target point of tetrodotoxin(TTX) on nerve system, and the possible consequence of this toxic to human.

No.	
1#	
2#	

Part V: Bonus **6*2=12 marks**

1. Draw a diagram of action potential,
   From the formula $I_{ions}=g_{ions}(V_m-E_{ion})$, **Label** the area on your action potential graph when the **Na^+ influx** current is the **Lowest.**
2. A neuron was set in V_m=0 mV, permeability to Na+ has been triggered to 0, **explain with reasons**, what will happen when the dynamic change of membrane potential in this neuron has reached equilibrium.

E_{ion} $P_K = P_{Na} \times 40.$

$V_m = E_k = -80\,mV.$

